CHIRAG BYANJANKAR, E.I.T.

cbyanjankar99@gmail.com | (+1) 940-453-5212 | LinkedIn
GitHub | Website

Professional Summary

Mechanical Engineering Ph.D. Student specializing in thermal management, spacecraft environmental control systems, and additive manufacturing. Proven expertise in experimental design, CFD analysis, and AI-enhanced manufacturing. Published researcher with NASA-funded experience.

Education

Ph.D. in Mechanical and Energy Engineering

University of North Texas, Denton, TX — Aug 2025 – Present

Advisor: Prof. H. Bostanci

M.S. in Mechanical and Energy Engineering

University of North Texas, Denton, TX — Aug 2022 – Dec 2024

CGPA: 4.0/4.0 | Thesis: Vortex Phase Separator-based Spacecraft Cabin Air Humidity Control Subsystem for CO_2 Removal

B.S. in Mechanical and Energy Engineering

Kathmandu University, Kathmandu, Nepal — Aug 2016 – Dec 2020

CGPA: 3.23/4.0

Research Experience

Graduate Research Assistant – Thermal Management Lab

University of North Texas, Denton, TX — May 2023 – Present

- Designed and prototyped Vapor Phase Separator (VPS) for spacecraft humidity removal under NASA X-Hab 2023 Challenge
- Achieved 90% air humidity removal capability for deep space applications
- Developed LabVIEW data acquisition software for pressure, temperature, and humidity measurement
- Co-authored 2 peer-reviewed papers at ICES 2023 and 2024 conferences

Graduate Research Assistant – AI Lab

University of North Texas, Denton, TX — May 2023 – Aug 2023

- Fine-tuned SAM (Segment Anything Model) for defect detection in additive manufacturing
- DEVCOM ARL-funded program on 3D printing defect detection
- Integrated YOLO object detection with SAM adapters for enhanced localization

Professional Experience

Mechanical Engineer

Microtech M&E Pvt. Ltd., Kathmandu, Nepal — Oct 2020 – Jun 2022, Oct 2024 – Aug 2025

- Led HVAC, fire-fighting, and plumbing systems for Sheraton Hotel Kathmandu
- Directed multi-disciplinary teams in HVAC system design and NFPA compliance
- Reviewed technical submittals for accuracy and specification adherence

Graduate Teaching Assistant

Department of Mechanical Engineering, University of North Texas Sep 2023 – May 2024, Aug 2025 – Present

- Established lab equipment and manuals for Thermal Sciences and Refrigeration courses
- Taught 60+ students eQUEST building energy simulation
- Mentored undergraduate solar decathlon team on EnergyPlus HVAC design

Publications

Byanjankar, C., Bostanci, H., Kurwitz, C., Belancik, G. (2024). Design, Development, and Initial Testing of Microgravity Vortex Phase Separator-Based Spacecraft Cabin Air Humidity Control Subsystem for CO_2 Removal. *53rd International Conference on Environmental Systems (ICES 2024)*, Louisville, Kentucky.

Byanjankar, C., Sarvadi, A., Bostanci, H., Kurwitz, C., Belancik, G. (2023). Preliminary Investigation of Vortex Phase Separator-Based Spacecraft Cabin Air Dehumidification Subsystem for CO2 Removal. *52nd International Conference on Environmental Systems (ICES 2023)*, Calgary, Canada.

Technical Skills

CAD & Design: SolidWorks, AutoCAD, CATIA
CFD & Simulation: ANSYS Fluent, COMSOL, EnergyPlus, eQUEST, OpenFOAM
Data Acquisition: LabVIEW, Arduino, RaspberryPi
Programming: Python, MATLAB, C/C++
Machine Learning: SAM, YOLO, PyTorch, TensorFlow
Tools: Git/GitHub, Tecplot, HOMER Pro, Microsoft Office

Certifications and Awards

- Six Sigma Green Belt (Sep 2024)
- University of North Texas GTA Award (2023–2024)

Professional Memberships

- International Conference on Environmental Systems (ICES) Attendee